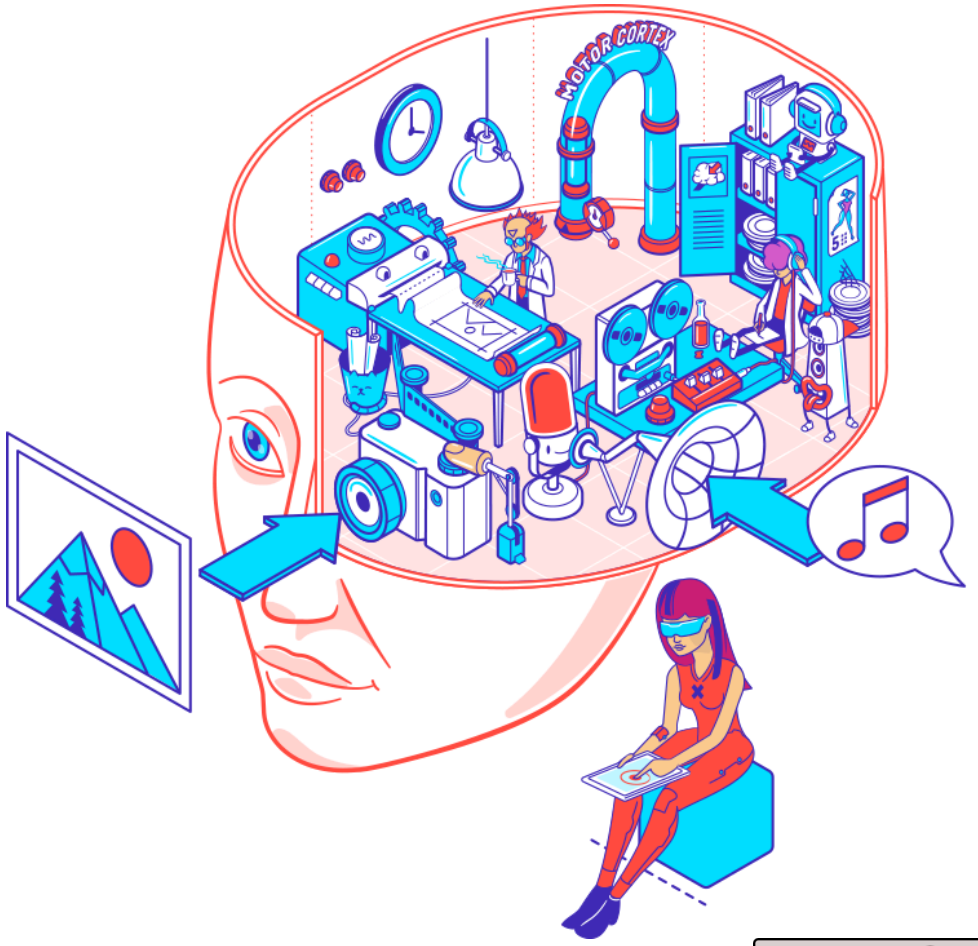
HAW HAMBURG

MEDIA PROGRAMMING

UNIT 0: ORGANISATION



Prof. Dr. Eike Langbehn

Department of Media Technology
Faculty of Design, Media and Information
Hamburg University of Applied Sciences

CC BY

INTRODUCTION ROUND

UNIT 0: ORGANISATION



EIKE LANGBEHN

- **Since 3/2021:** Professor for Media Informatics, HAW Hamburg
- **Since 1/2020:** Creative Director & Co-Founder, Curvature Games GmbH
- **1/2015 - 12/2019:** Wissenschaftlicher Mitarbeiter & Doktorand, Uni Hamburg
- **12/2014:** M.Sc. in Informatics, Uni Hamburg
- **03/2012 - 12/2013:** Software-Developer Unity3D, XYRALITY GmbH
- **09/2011:** B.Sc. in Informatics, CvO Uni Oldenburg



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WHO ARE YOU?



CLASS CONTENT

UNIT 0: ORGANISATION



LEARNING OUTCOME

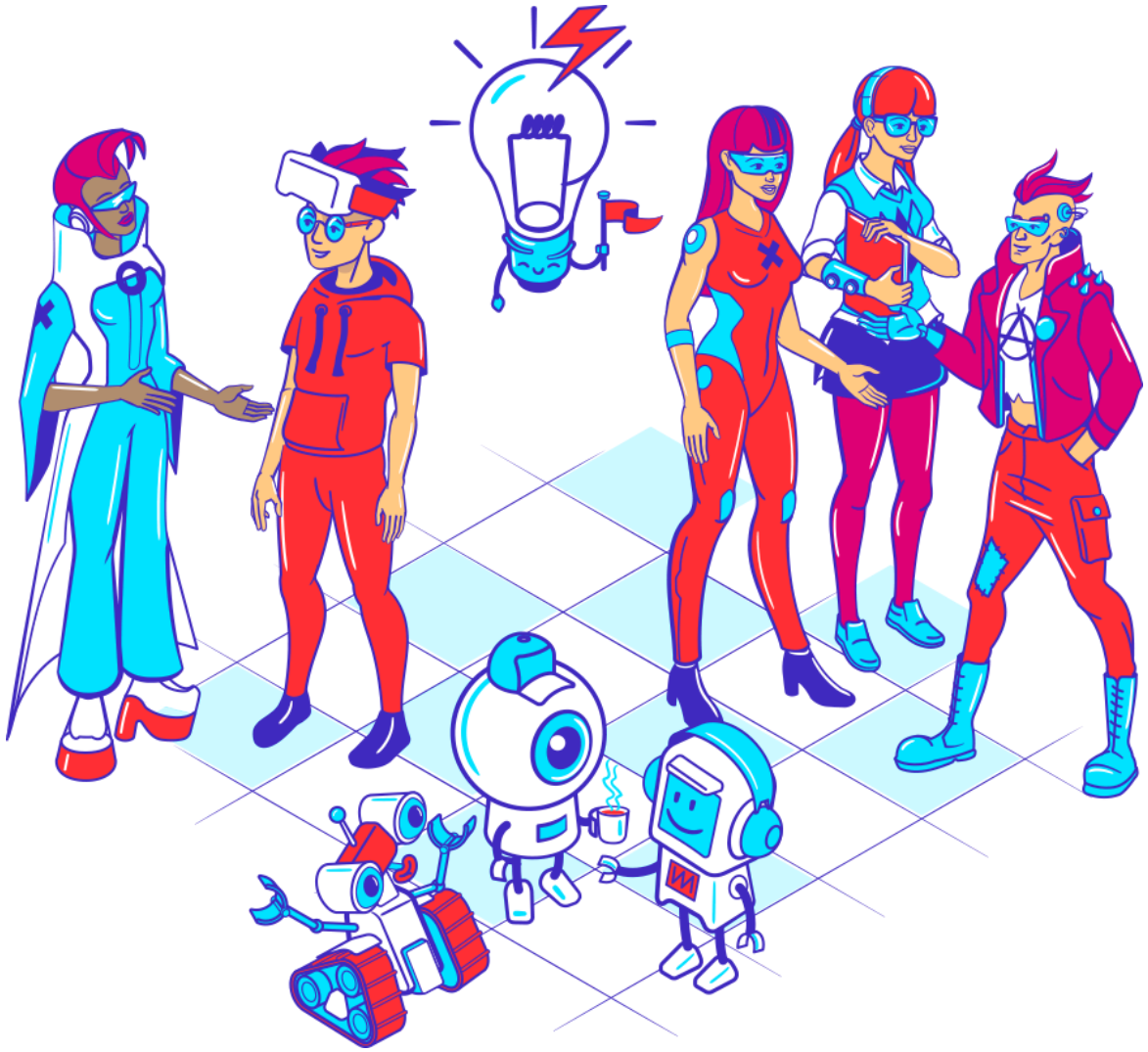
- Die Studierenden lernen die Grundlagen digitaler Medien und wenden programmiertechnische Methoden an um digitale Medien zu **verarbeiten**, zu **speichern**, zu **manipulieren** und zu **erstellen**.

- Die Studierenden
  • kennen verbreitete Formate digitaler Medien
  • können grundlegende Algorithmen zur Verarbeitung und Manipulation von digitalen Medien implementieren

LEARNING OUTCOME

- Erwerb von grundlegenden Kenntnissen für die Entwicklung von Grafik-basierten Anwendungen.
- Die Studierenden können Methoden anwenden um interaktive Inhalte digitaler Medien zu erstellen.

WHAT IS DIGITAL MEDIA?



WHAT IS DIGITAL MEDIA?

Digital Media refers to any media that are encoded in machine-readable formats. It can be created, viewed, distributed, modified and preserved on digital electronics devices.

- Text & Documents
- Audio (MP3, AAC)
- Video (MP4, Streaming)
- Images (JPEG, PNG, SVG)
- Interactive Environments

ENCODING & COMPRESSION

1. ENCODING (ASCII)

Turning text into binary **0s and 1s**.

```
"HEY"
H = 72 → 01001000
E = 69 → 01000101
Y = 89 → 01011001
```

2. COMPRESSION (RLE)

Reducing size by **removing redundancy**.

```
"AAABBBBAA"
Uncompressed: 9 chars
Compressed: 3A4B2A (6 chars)
```

ENCODING & COMPRESSION

TASK 1: ENCODING

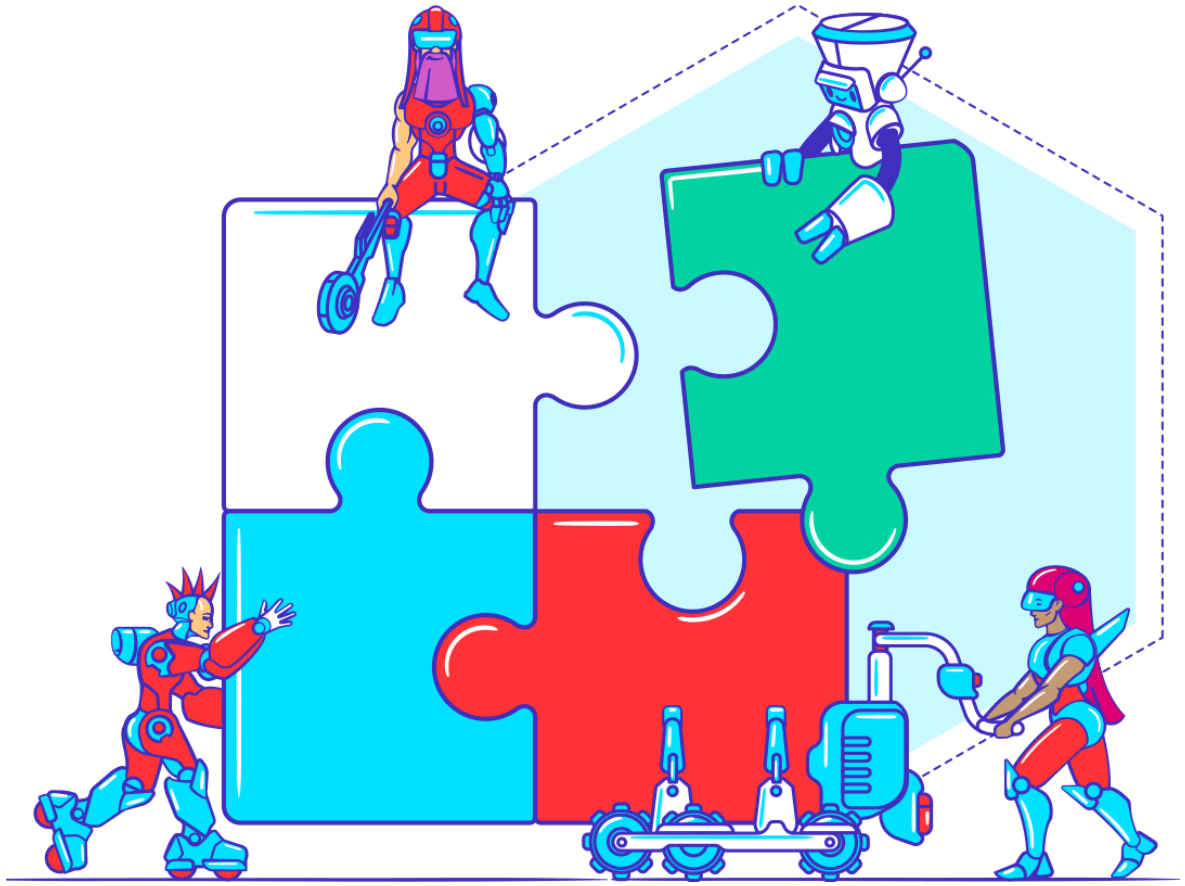
Encode the word **"CAB"** into binary using ASCII values

TASK 2: COMPRESSION

Apply **Run-Length Encoding (RLE)** to the following string:

BBBAACCC

How many characters were saved?



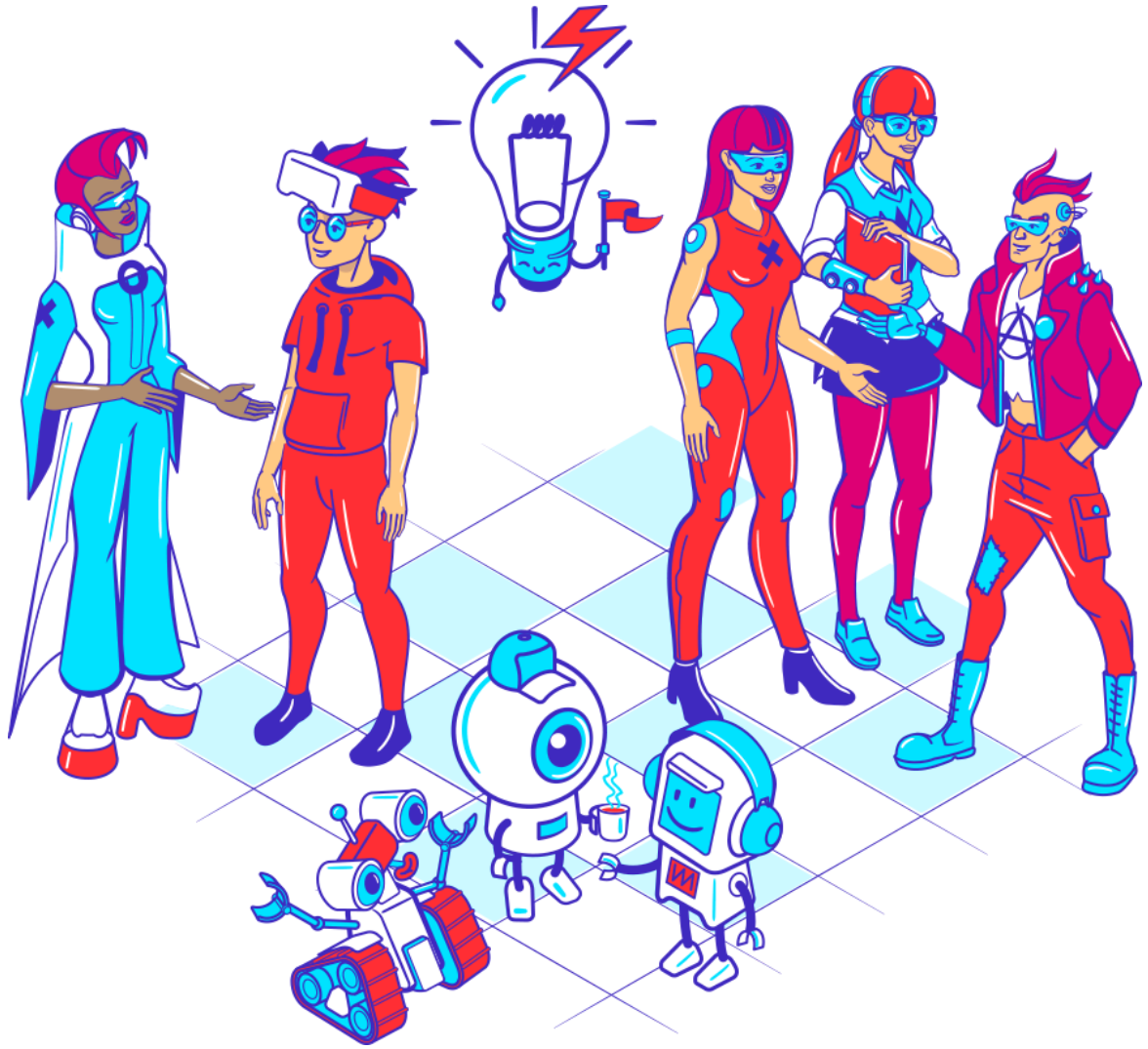
MEDIA PROGRAMMING

Media Programming involves writing software to create, manipulate, or display digital media. It is the logic that powers **social media filters, games, generative art, real-time video and audio streaming.**

GROUP DISCUSSION

- What is a piece of software you use daily that relies on **media programming**?

- How does **code** change the way we experience a simple image or sound?

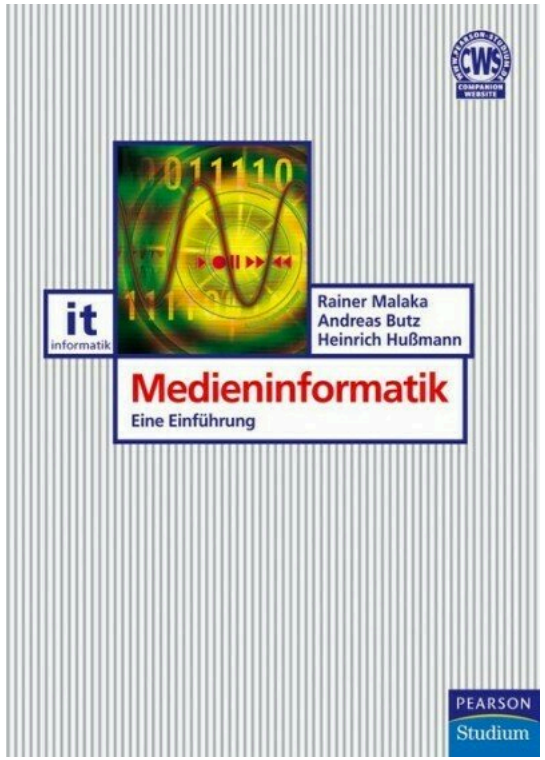


AGENDA:

GRAPHICS PROGRAMMING

1. **Introduction**: Visual Perception, Light, Color
2. **From Java to Python**
3. **Raster Graphics**: Encoding, Compression, Filter
4. **Algorithmic Graphics Generation**
5. **Vector Graphics**: Encoding, Transformation, Rendering
6. **Interactive Graphics**: Godot Engine
7. **Animations**: Keyframe, Simulation

LITERATURE


it
informatik
Rainer Malaka
Andreas Butz
Heinrich Hußmann
Medieninformatik
Eine Einführung
PEARSON
Studium

CLASS STRUCTURE

UNIT 0: ORGANISATION

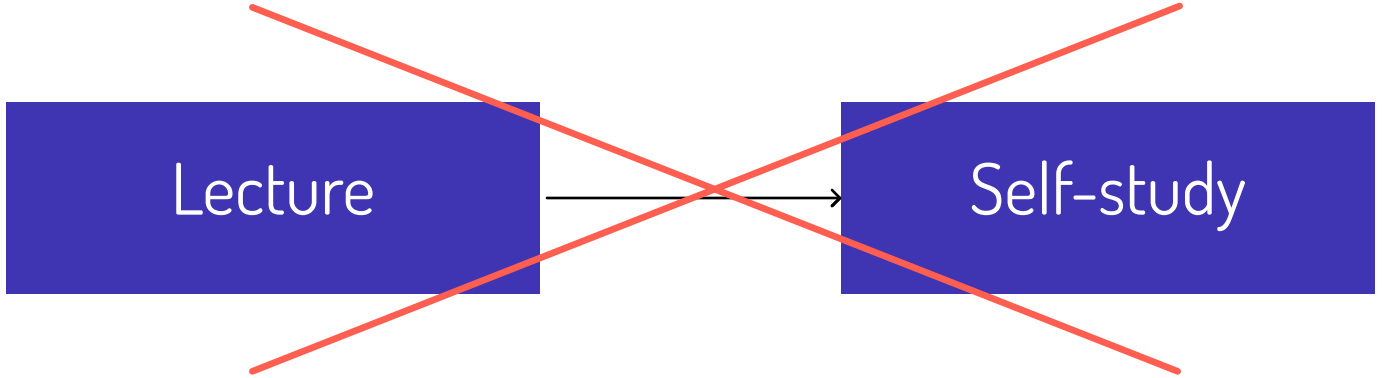


STRUCTURE

Zeitplan

Wann?	Was?	Wer?	Wo?
Dienstag, 14.04.2026, 8.15h - 11.45h	Graphics Programming	Eike Langbehn	Online in Teams
Dienstag, 21.04.2026, 8.30h - 11.45h	AV Programming	Eva Wilk	Finkenau - E62
Dienstag, 28.04.2026, 8.15h - 11.45h	Graphics Programming	Eike Langbehn	Online in Teams
Dienstag, 05.05.2026, 8.30h - 11.45h	AV Programming	Eva Wilk	Finkenau - E62
Dienstag, 12.05.2026, 8.15h - 11.45h	Graphics Programming	Eike Langbehn	Online in Teams
Dienstag, 19.05.2026, 8.30h - 11.45h	AV Programming	Eva Wilk	Finkenau - E62
Dienstag, 26.05.2026, 8.15h - 11.45h	Graphics Programming	Eike Langbehn	Online in Teams
Dienstag, 02.06.2026, 8.30h - 11.45h	AV Programming	Eva Wilk	Finkenau - E62
Dienstag, 09.06.2026, 8.15h - 11.45h	Graphics Programming	Eike Langbehn	Online in Teams
Dienstag, 16.06.2026, 8.30h - 11.45h	AV Programming	Eva Wilk	Finkenau - E62
Dienstag, 23.06.2026, 8.15h - 11.45h	Graphics Programming	Eike Langbehn	Online in Teams
Dienstag, 30.06.2026, 8.30h - 11.45h	AV Programming	Eva Wilk	Finkenau - E62
Dienstag, 07.07.2026, 8.15h - 11.45h	Graphics Programming	Eike Langbehn	Online in Teams
Dienstag, 14.07.2026, 8.30h - 11.45h	AV Programming	Eva Wilk	Finkenau - E62

PROCEDURE


Lecture
Self-study


between sessions
live
Self-study
Lecture (Plenary, exercises)

PROCEDURE

- Materials will be available in moodle after each session
  - slides, videos, links, questionnaires/quizzes
- Self-study until next session
- Interactive live sessions
  - tasks, exercises
  - questions, discussions

MATERIALS & DISCUSSIONS

Moodle: https://moodle.haw-hamburg.de/course/view.php?id=10534

Language: English (materials), German (spoken)

EXAMINATION

- Eine gemeinsame Klausur für das gesamte Modul
- Termin in der zweiten Prüfungsphase im Oktober zu Beginn des WS26/27
- Während des Semesters 2 Tests
  - Jeder geht mit 10% in Gesamtnote ein
  - 1. Test: nach der Hälfte
  - 2. Test: am/gegen Ende

QUESTIONS?

UNIT 0: ORGANISATION

